

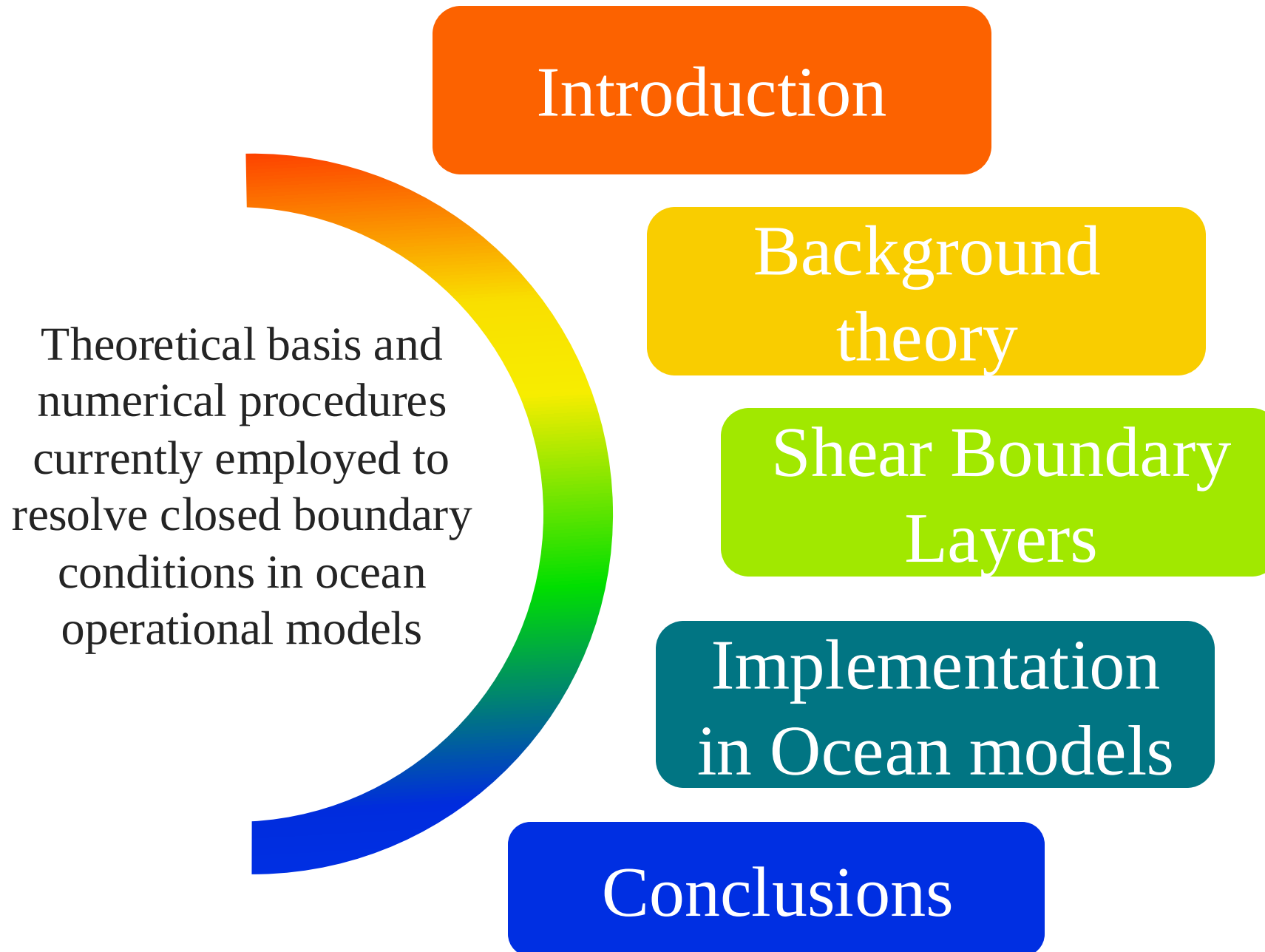


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Theoretical basis and numerical procedures currently employed to resolve closed boundary conditions in ocean operational models

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Introduction

$$\frac{\partial \rho C}{\partial t} + \nabla \cdot (\rho \mathbf{u} C) = \nabla \cdot (\rho k_c \nabla C) \quad \rho = f(T, S, p)$$

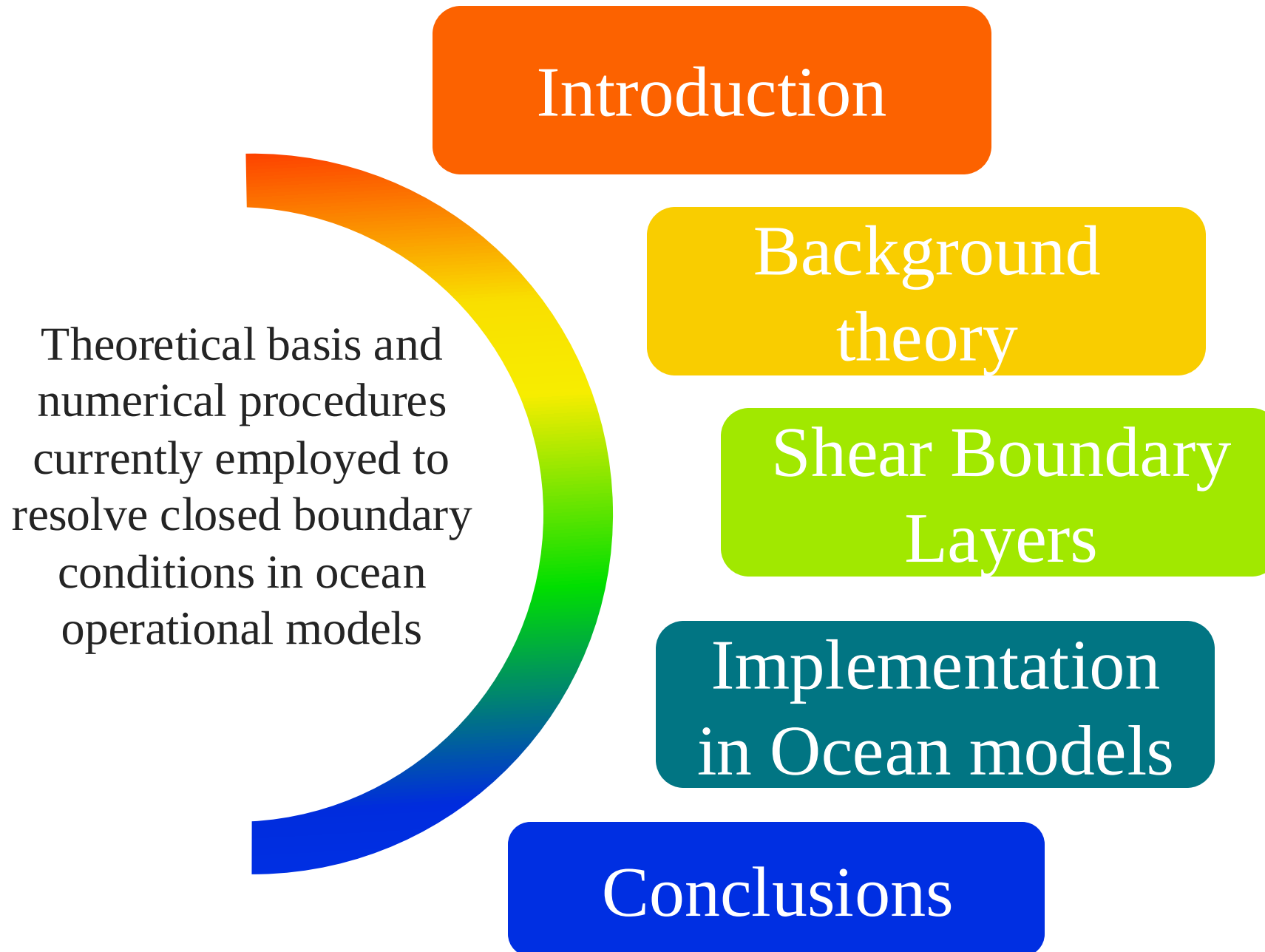
$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) + 2\boldsymbol{\Omega} \times (\rho \mathbf{u}) = -\nabla p - \hat{\mathbf{k}} g \rho + \nabla \cdot (\mu \nabla \mathbf{u}) + \frac{1}{3} \nabla (\mu \nabla \cdot \mathbf{u})$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

Navier-Stokes equations are partial differential equations in time and space, so they require the imposition of appropriate initial and boundary conditions to find a specific solution.

The boundary conditions refer to the 'physical limits' of the fluid under consideration, and they are specific for every problem of interest.

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Background theory

On the boundary of the basin the normal velocity must be specified.

If the basin has rigid lateral boundaries and is closed, the normal velocity must vanish on the boundary. Thus, on the boundary:

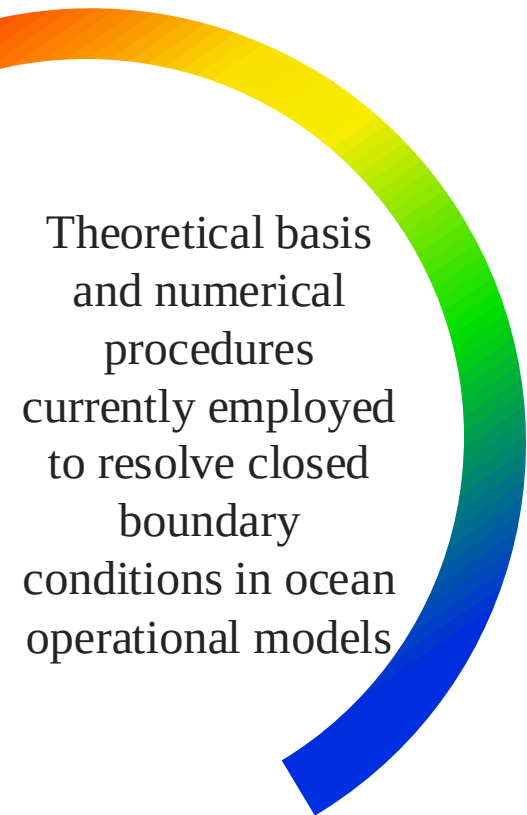
$$\mathbf{u}_h \cdot \mathbf{n} = 0$$

where $\mathbf{n}$ is a unit vector normal to the boundary and $\mathbf{u}$ is the horizontal velocity vector.

Since:

$$\mathbf{u} = \hat{\mathbf{k}} \times \nabla \psi$$

this is equivalent to the condition that ψ be constant on the boundary.



The presence of viscous diffusion implies the need for an additional dynamic boundary condition on the walls of the basin.

If we interpret conservation of momentum equation (also in vorticity form) as applying to all scales of motion, the standard condition of no slip on the boundary would be required.

This would be:

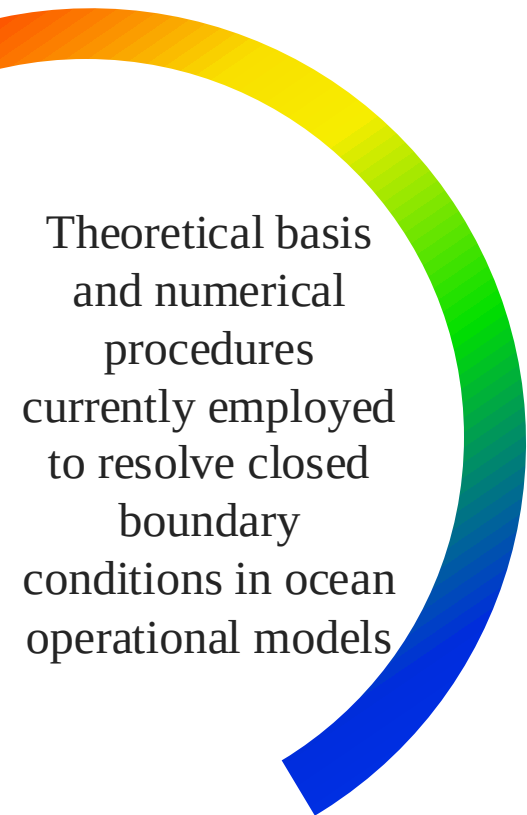
$$\mathbf{u}_h \cdot \mathbf{t} = 0$$

No SLIP

where $\mathbf{t}$ is a unit vector tangent to the boundary. In terms of ψ this becomes:

$$\mathbf{n} \cdot \nabla \psi = 0$$

No SLIP



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It is not clear, however, that is the correct boundary condition when our equations explicitly resolve only the large-scale motion.

The no-slip condition is a reflection of the interaction of the fluid with a solid boundary on the microscopic level, and there is no reason why motion on scales of the order of several kilometers should mimic that interaction.

We might even argue instead that the small-scale processes that occur right against the wall buffer the large-scale flow and allow it to slip smoothly along the wall. A condition of no tangential stress for the large-scale flow might be equally logical then.

$$\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} = 0$$

$$\zeta = \nabla^2 \psi = 0$$

Free SLIP



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On the other hand, if the lateral diffusion term is considered to be a parameterization of small-scale mixing of vorticity by an unresolved eddy field, we could argue that we are actually specifying a relation between the eddy flux of total vorticity and the gradient of the large-scale vorticity field, i.e.:

$$-\overline{u'\zeta'} = A_H \nabla(\zeta + \beta y)$$

At a solid boundary the normal component of the eddy flux of vorticity must vanish since the eddy normal velocity must vanish there. This would imply that:

$$\hat{n} \cdot \nabla(\nabla^2 \psi) + \beta \hat{n} \cdot \hat{j} = 0$$

Superslip/hyperslip

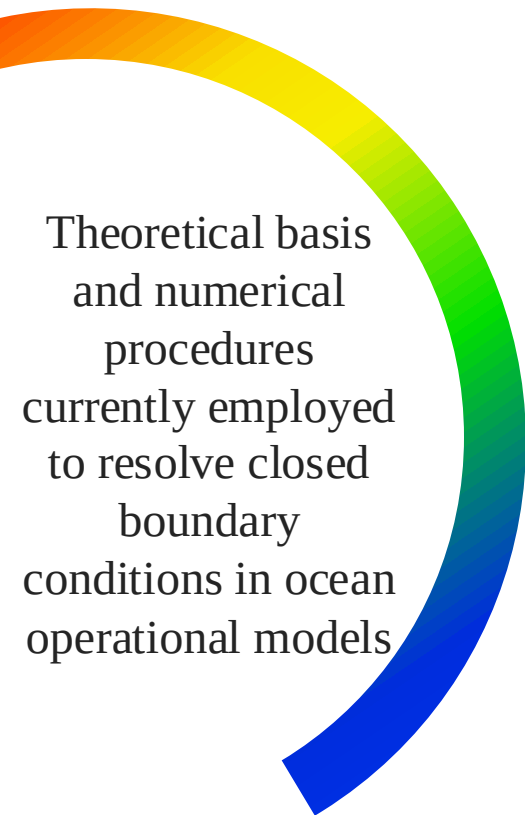
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Which boundary condition should we apply?

The choice makes a great difference.

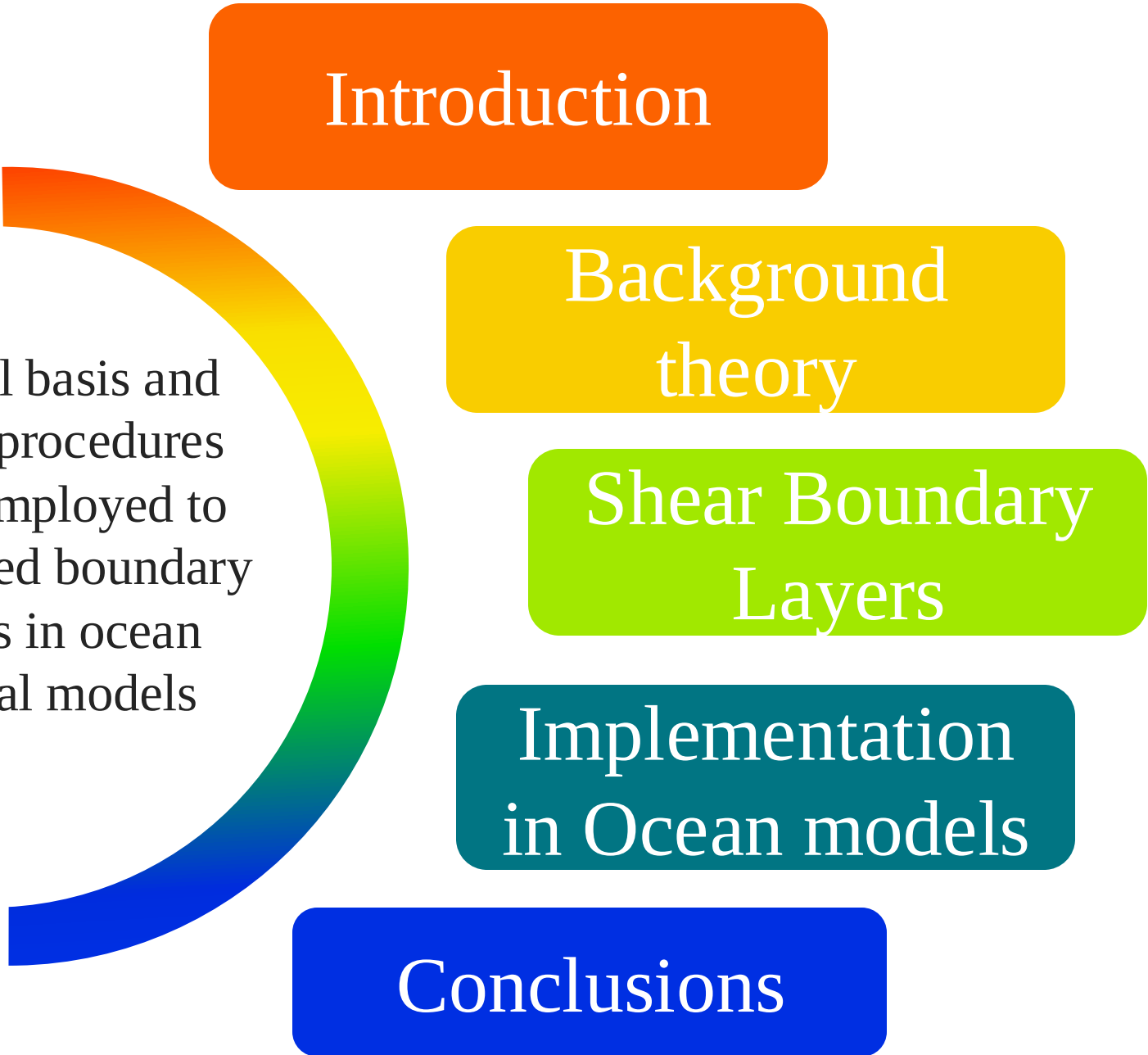
Some patterns of the circulation depend on the type and the magnitude of the dissipation (lateral mixing or bottom friction).

The application of different boundary conditions implies different choices of the dissipation mechanism which affects the way the fluid can contrive to balance the input of vorticity by the wind or other sources.



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Shear Boundary Layers

Atmospheric boundary layer dynamicists have been long faced with this issue when computing the momentum transfers between the lower boundary and the atmosphere, be it land or ocean.

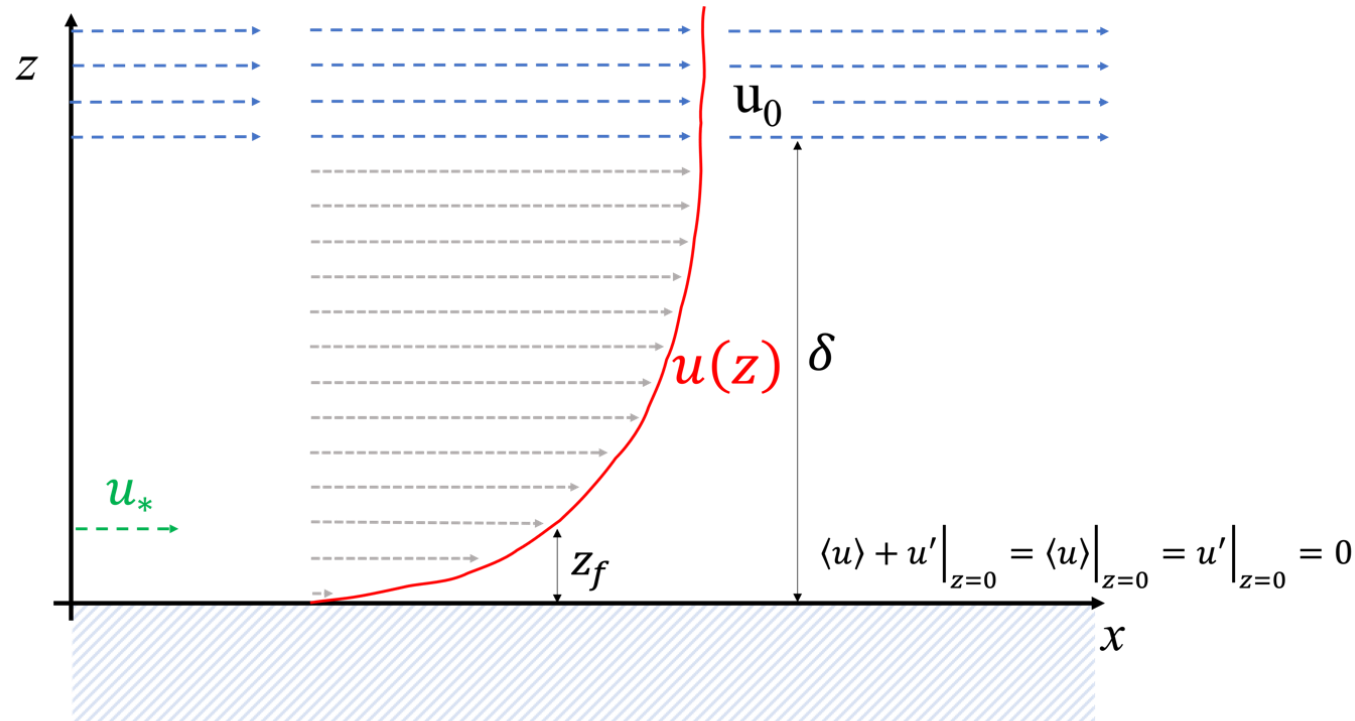
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Shear Boundary Layers

At fluid boundaries the only stress exerted directly on the wall is the viscous one. Away from the wall, instead, large velocity gradients generate turbulence through Reynolds stresses which are typically large compared to the viscous stress.

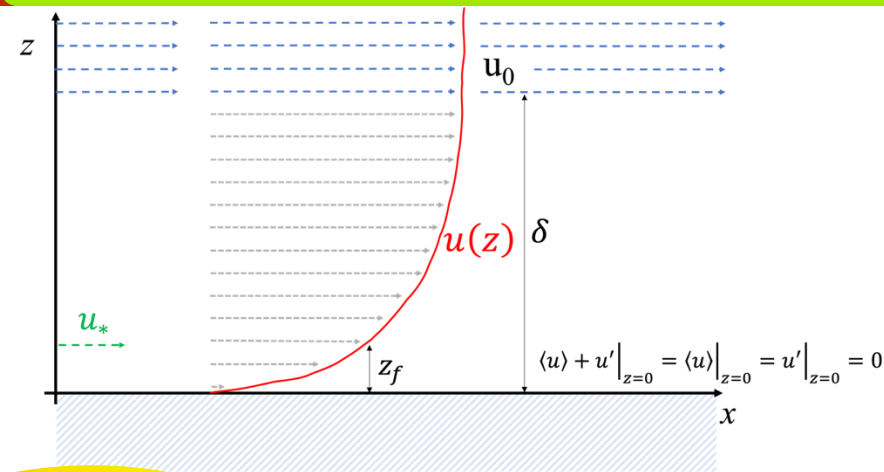
As a result, there is a viscous sublayer next to the wall, whose width is set by viscous forces, and a high Re boundary layer, whose thickness is controlled by the turbulent Reynolds stresses. These two layers are separated by transition sublayer.



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Shear Boundary Layers



Assumptions:

- a constant background flow u_0 independent of distances along and normal to the boundary.
- The flow is 2-dimensional ($\partial/\partial y = 0$).
- Averaged vertical velocities do not vary significantly along the direction normal to the boundary.
- Horizontal length scales are much larger than the size of the boundary layer. If L is the horizontal scale, we are assuming $\delta/L \ll 1$, so that for averaged variables we can neglect variations along the x direction compared to those in the vertical (i.e., $\partial/\partial x = 0$).
- Neglect the effect of rotation.

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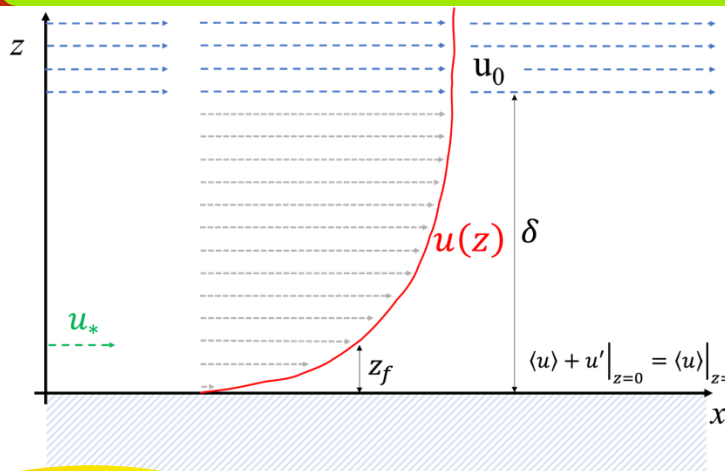
$$\rho \langle w \rangle \frac{\partial \langle u \rangle}{\partial z} = \frac{\partial}{\partial z} \left(\rho \nu \frac{\partial \langle u \rangle}{\partial z} - \rho \langle w' u' \rangle \right)$$

$$\frac{\partial \langle w \rangle}{\partial z} = 0$$

$$\frac{\partial}{\partial z} \left(\rho \nu \frac{\partial \langle u \rangle}{\partial z} - \rho \langle w' u' \rangle \right) = 0$$



Shear Boundary Layers



So, the stress τ , which is the sum of two terms:

$$\tau = \rho \nu \frac{\partial \langle u \rangle}{\partial z} - \rho \langle w' u' \rangle$$

will be dominated by the viscous term near the boundary: $\tau|_{z=0} = \rho \nu \frac{\partial \langle u \rangle}{\partial z} \Big|_{z=0}$

and by the turbulent velocity fluctuation “away” from the boundary:

$$\tau|_{z>z_f} = -\rho \langle w' u' \rangle|_{z>z_f}$$

We can define a characteristic velocity and length scales as

$$z_f = \frac{\nu}{u_*} \quad u_* = \sqrt{\tau/\rho}$$

where u_* is the friction velocity and z_f is the viscous length scale (the transition between the inertial and viscous sublayers).

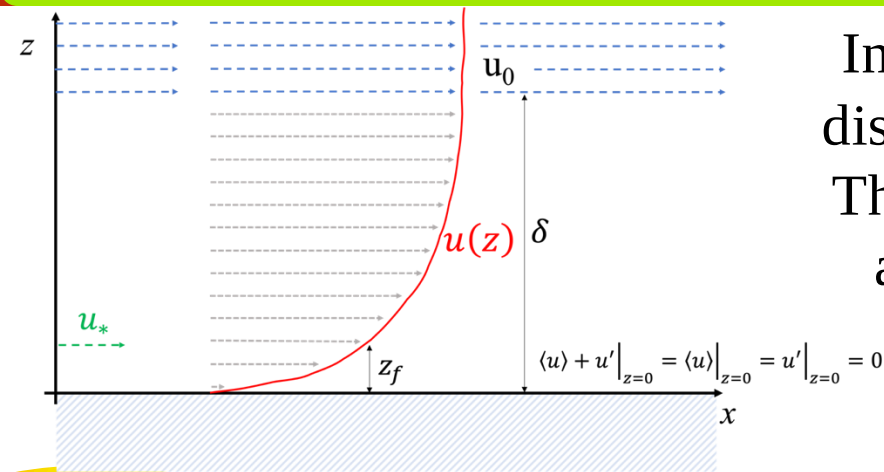
Based on the friction velocity, the viscous length scale, and the total depth of the boundary layer (δ), we can also form non-dimensional velocity and heights as:

$$u^+ = \frac{u}{u_*} \quad z^+ = \frac{z u_*}{\nu} \quad \eta = \left(\frac{z}{\delta} \right)$$

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Shear Boundary Layers



In the viscous, or inner, sublayer $z < z_f$, the velocity must depend on the distance from the boundary z , the friction velocity u^* , and the viscosity ν . The dimensional quantities involved in the viscous sublayer are u , z , τ , ρ and ν (not δ) but there are only 3 independent dimensions, so we can define 2 independent dimensionless groups, u^+ and z^+ .

We can write the expected relationship in non-dimensional form as:

$$u^+ = f_1(z^+)$$

Where f_1 , called **law of the wall**, is expected to be a universal function and should be able to satisfy the boundary condition.

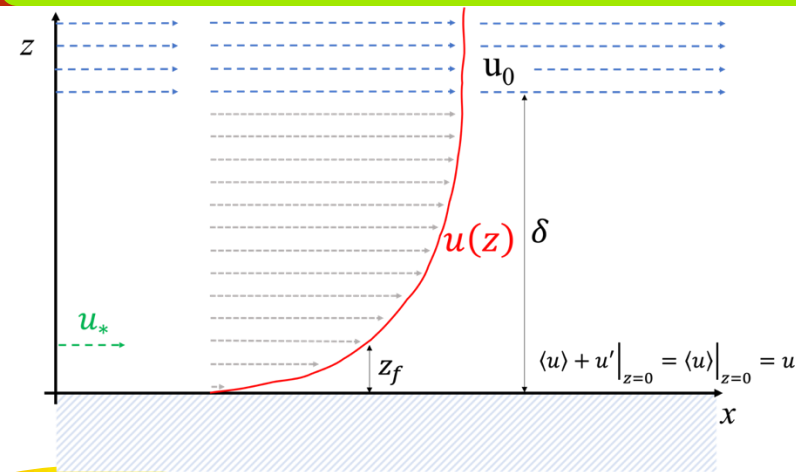
Outside the viscous sublayer, we can neglect viscosity, but we need to account for δ (the total depth of the boundary layer). In this sublayer the relation should be such that for $z \rightarrow \infty$ $u = u_0$, so that the relation is:

$$\frac{u_0 - u}{u_*} = f_2(\eta)$$

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Shear Boundary Layers



We have two different laws for u^+ . One applies close to the wall in the viscous sublayer and satisfies the no-slip condition. The other applies further away from the wall and is not guaranteed to satisfy the no-slip boundary condition at the wall. This indicates that a viscous sublayer with very steep gradients is required to satisfy the boundary conditions. The velocity doesn't jump from one scaling behavior to another - there is a transition region.

In this transition region we expect both the law of the wall (f_1) and the velocity defect law (f_2) to apply. In non-dimensional form we have:

$$u_0^+ - u^+ = f_2(\eta)$$

$$u^+ = f_1(z^+)$$

We can introduce the dimensionless parameter $\delta^+ = \delta u_* / \nu$ so that $z^+ = \eta \delta^+$ and $u_0^+(\delta^+)$. The latter because u_0^+ only depends on u_* which is also the only variable in δ^+ (being δ and ν constant). So that, substituting we get:

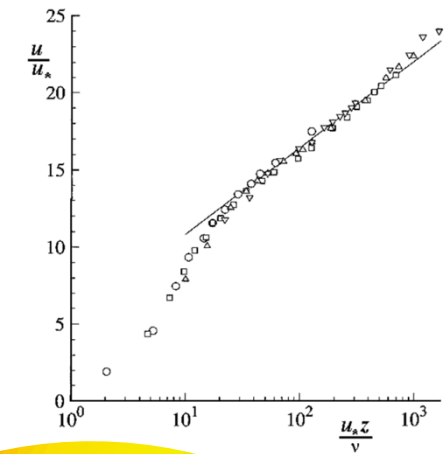
$$u_0^+(\delta^+) = f_1(\eta \delta^+) + f_2(\eta)$$

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Shear Boundary Layers



For a function f_1 of the products $\eta\delta^+$ to be the sum of separate functions η and δ^+ , f_1 must be logarithmic. Differentiate wrt δ^+

$$u_0^+(\delta^+) = f_1(\eta\delta^+) + f_2(\eta)$$

$$u_0^+(\delta^+) = \eta f_1'(\eta\delta^+) + 0$$

Differentiate wrt η

$$0 = f_1'(\eta\delta^+) + \eta\delta^+ f_1''(\eta\delta^+) \rightarrow 0 = f_1'(z^+) + z^+ f_1''(z^+) \rightarrow 0 = \frac{d}{dz^+} \left(z^+ \frac{df_1}{dz^+} \right)$$

Hence $z^+ \frac{df_1}{dz^+} = \text{constant}$

The observational determination of the constant in the absence of rotation has been repeated countless times. The curve behaves as a straight line over a range of two orders of magnitude (from $u_* z / \nu$ between 10^1 and 10^3). This constant is conventionally written as $1/k$ where $k = 0.41$ is called the *von Kármán* constant

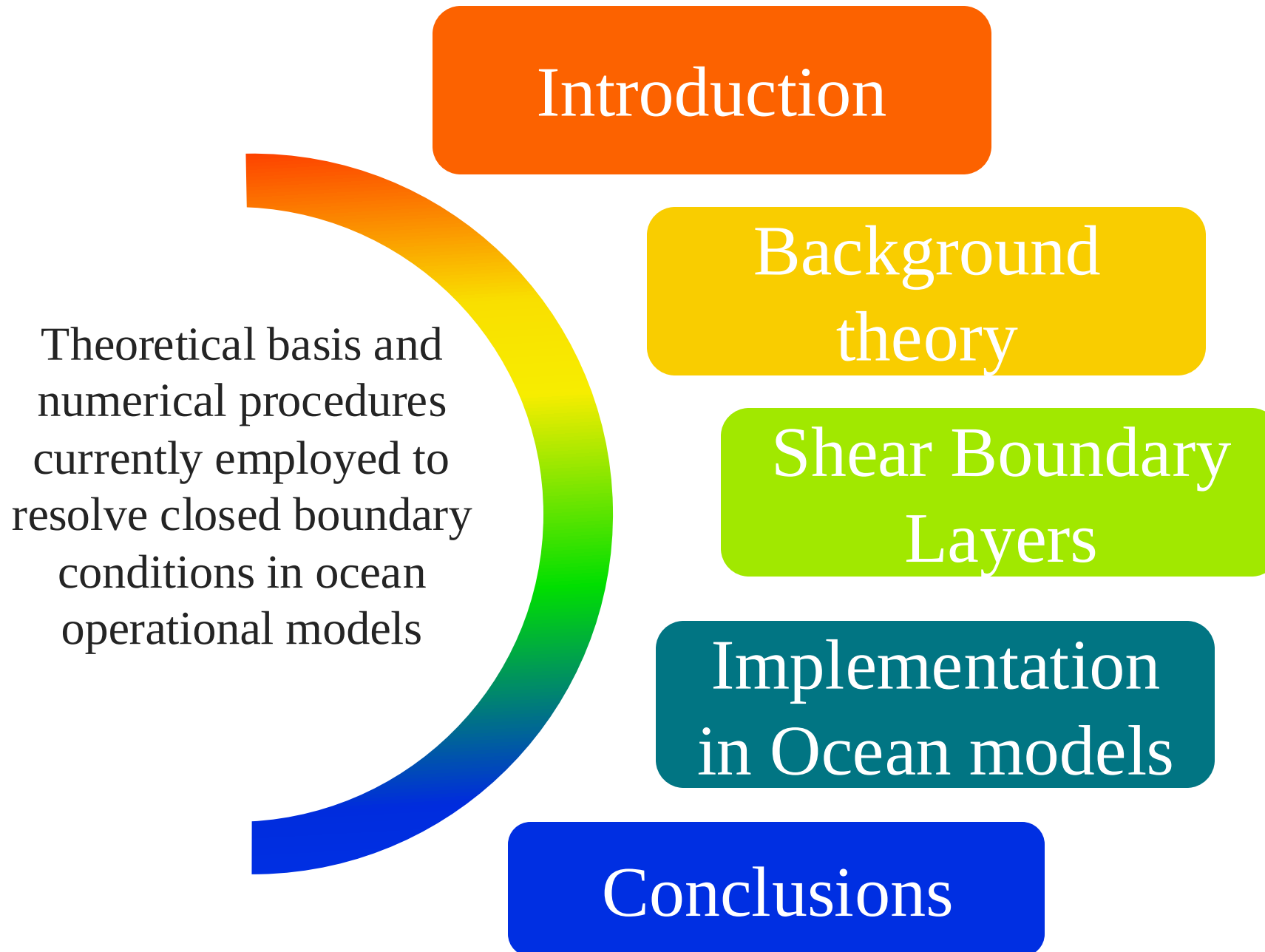
$$f_1 = \frac{1}{k} \ln(z^+) + B$$

$$u^+ = \frac{1}{k} \ln(z^+) + B$$



The region where this applies is known as the logarithmic layer.

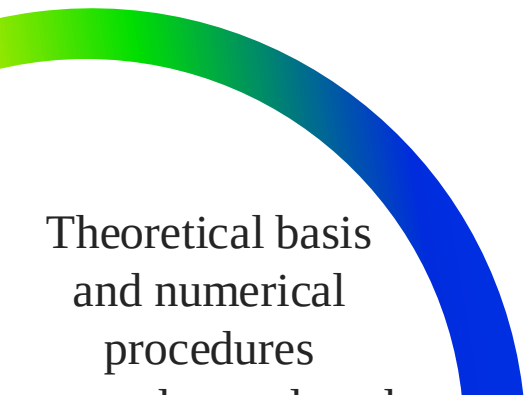
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Implementation in Ocean models

All the options are available in ocean models:

- No slip
- Free slip
- Partial slip
- Strong slip

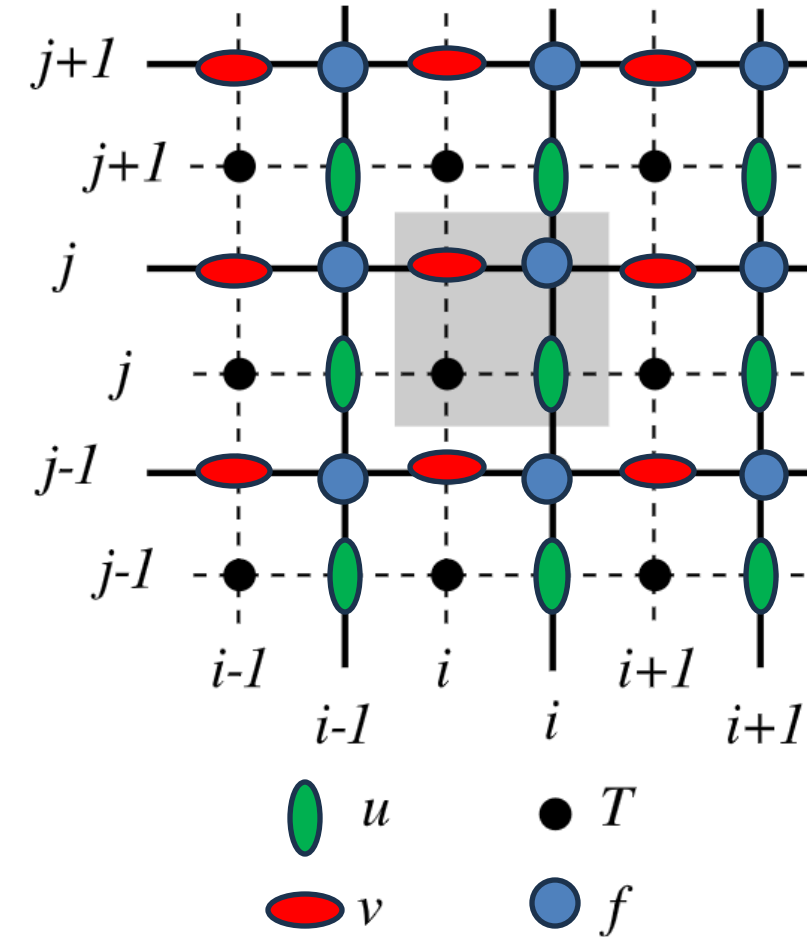
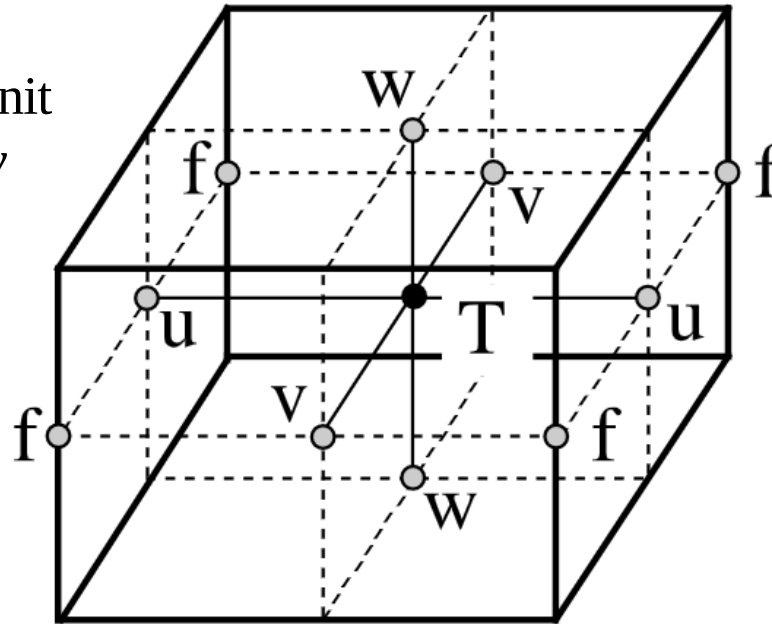


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Implementation in Ocean models

Arrangement of variables in the unit cell. T (tracers); f vorticity; u, v, w velocity



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Horizontal integer
indexing. The dashed
area indicates the cell
which variables have
the same i and j index



Implementation in Ocean models

An efficient way to work over the whole domain while specifying the boundary conditions, is to use multiplication by mask arrays in the computation. A mask array is a matrix whose elements are 1 in the ocean domain and 0 elsewhere.

A simple multiplication of a variable by its own mask ensures that it will remain zero over land areas.

Since most of the boundary conditions consist of a zero flux across the solid boundaries, they can be simply applied by multiplying variables by the correct mask arrays, i.e. the mask array of the grid point where the flux is evaluated.

For example, the heat flux in the x-direction

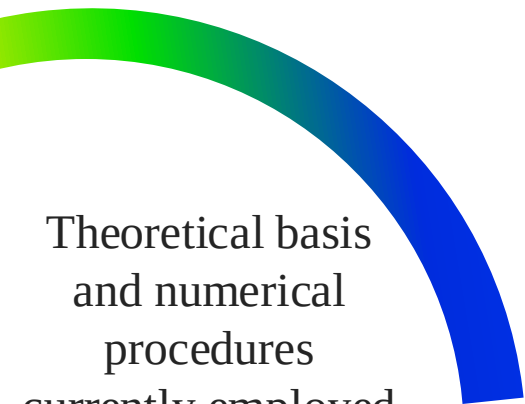
$$A_H \frac{\partial T}{\partial x} = \frac{T^{x+1} - T^x}{\Delta x} \cdot \text{mask_u}$$



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Implementation in Ocean models

For the momentum the situation is a bit more complex as two boundary conditions must be provided along the coast (for the normal and tangential velocities). The boundary of the ocean in the C-grid is defined by the velocity-faces. The lateral boundary (a coastline) is made of segments joining *f*-points, and normal velocity points are located between two *f*-points.



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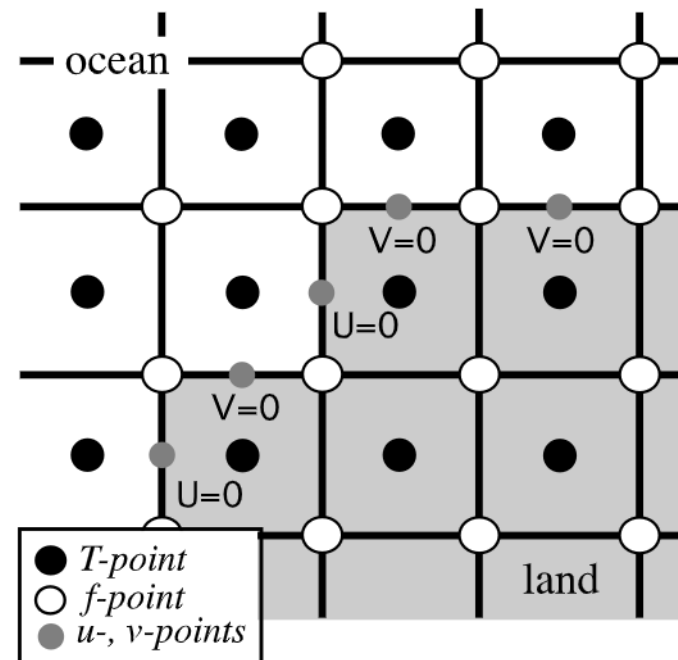
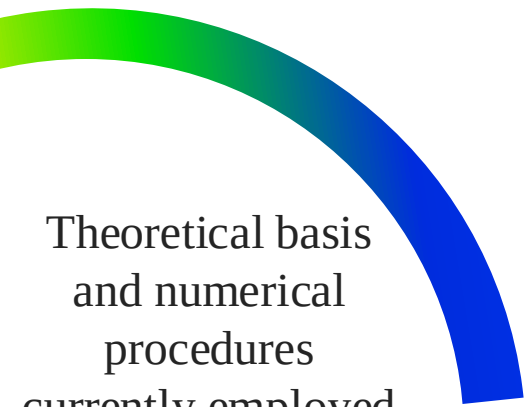


Figure 7.1.: Lateral boundary (thick line) at T-level. The velocity normal to the boundary is set to zero.



Implementation in Ocean models

The boundary condition on the normal velocity (no flux through solid boundaries) can be easily implemented using the mask system. The boundary condition on the tangential velocity requires a more specific treatment. This boundary condition influences the relative vorticity and momentum diffusive trends and is required in order to compute the vorticity at the coast.



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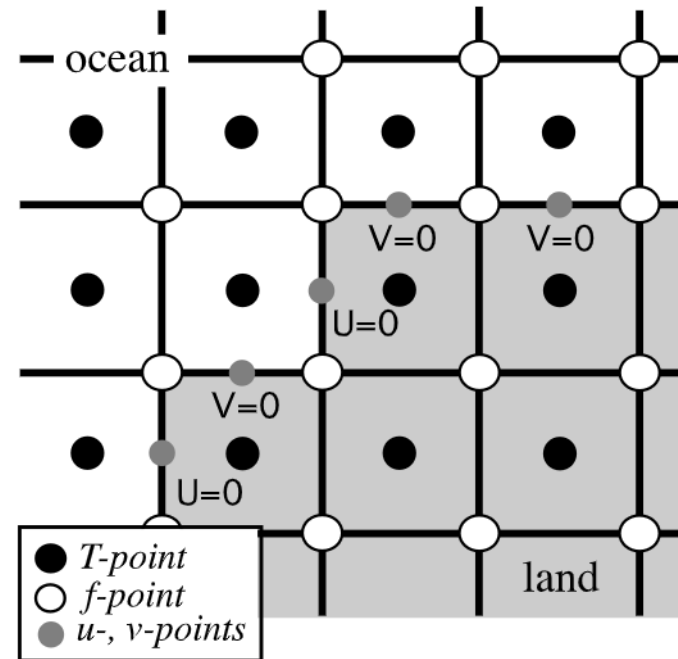


Figure 7.1.: Lateral boundary (thick line) at T-level. The velocity normal to the boundary is set to zero.

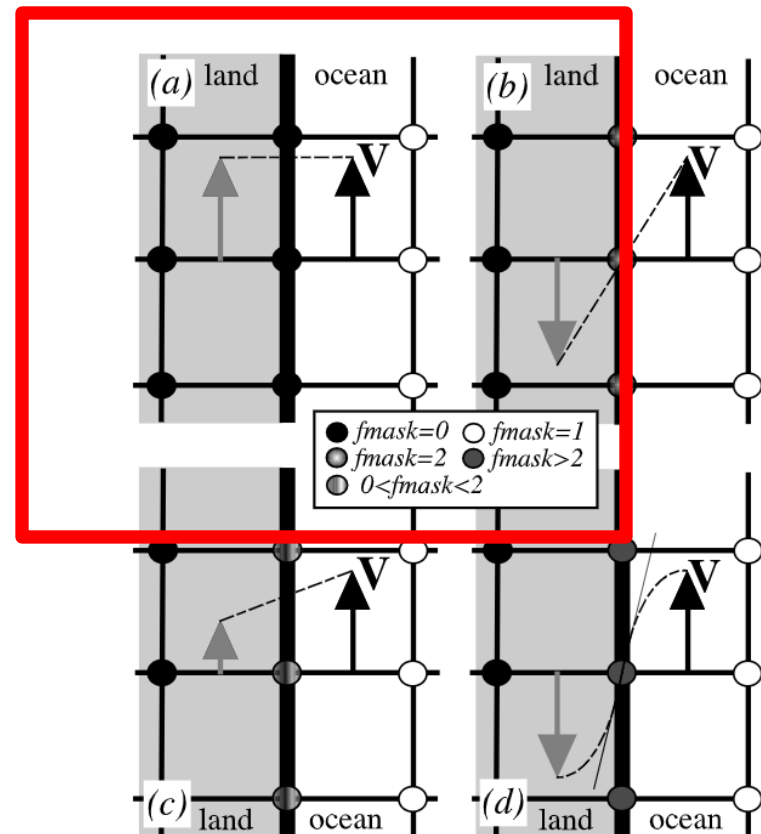


Implementation in Ocean models

Free-slip boundary condition ($mask-f=0$)

The tangential velocity at the coastline is equal to the offshore velocity, i.e. the normal derivative of the tangential velocity is zero at the coast, so the vorticity: $mask-f$ array is set to zero inside the land and just at the coast.

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$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - v^x}{\Delta x} \cdot mask_f$$

$$mask_f = 0$$

Figure 7.2.: Lateral boundary conditions (a) free-slip ($rn_shlat=0$); (b) no-slip ($rn_shlat=2$); (c) "partial" free-slip ($0 < rn_shlat < 2$) and (d) "strong" no-slip ($2 < rn_shlat$). Implied "ghost" velocity inside land area is display in grey.

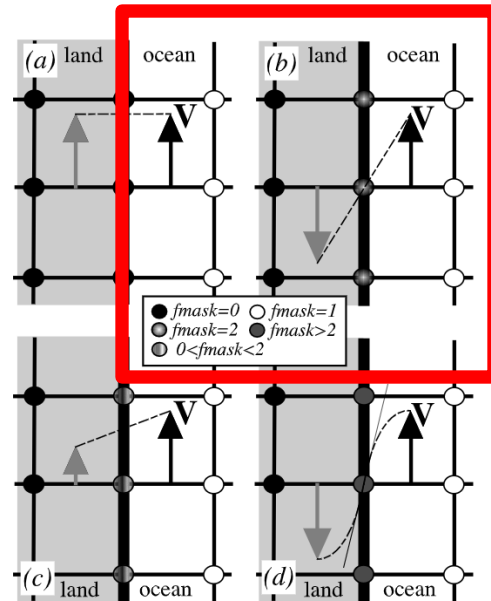


Implementation in Ocean models

No-slip boundary condition ($mask-f=2$)

the tangential velocity vanishes at the coastline. The normal derivative is evaluated as if the velocities at the closest land velocity grid point and the closest ocean velocity grid point were of the same magnitude but in the opposite direction. Setting the $mask-f$ array to 2 along the coastline provides a vorticity field computed with the no-slip boundary condition, simply by multiplying it by the $mask-f$

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$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - v^x}{\Delta x}$$

$$v^{x+1} = -v^x$$

$$\frac{\partial v}{\partial x} = 2 \frac{v^{x+1}}{\Delta x}$$

$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - \cancel{v^x}}{\Delta x} \cdot mask_f$$

$$mask_f = 2$$

Figure 7.2.: Lateral boundary conditions (a) free-slip ($rn_shlat=0$); (b) no-slip ($rn_shlat=2$); (c) "partial" free-slip ($0 < rn_shlat < 2$) and (d) "strong" no-slip ($2 < rn_shlat$). Implied "ghost" velocity inside land area is display in grey.

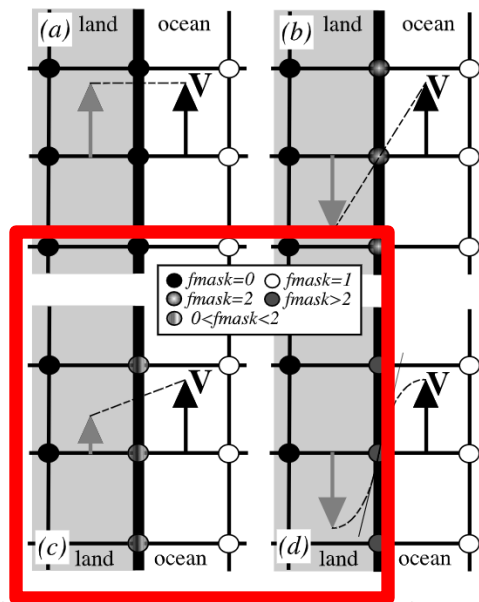


Implementation in Ocean models

”partial” free-slip boundary condition ($0 < mask_f < 2$)

the tangential velocity at the coastline is smaller than the offshore velocity, i.e. there is a lateral friction but not strong enough to make the tangential velocity at the coast vanish. This can be selected by providing a value of $mask_f$ in between 0 and 2 .

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$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - v^x}{\Delta x}$$

$$v^x = \frac{v^{x+1}}{\alpha}$$

$$\frac{\partial v}{\partial x} = \beta \frac{v^{x+1}}{\Delta x}$$

$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - \cancel{v^x}}{\Delta x} \cdot mask_f$$

$$0 < mask_f < 2$$

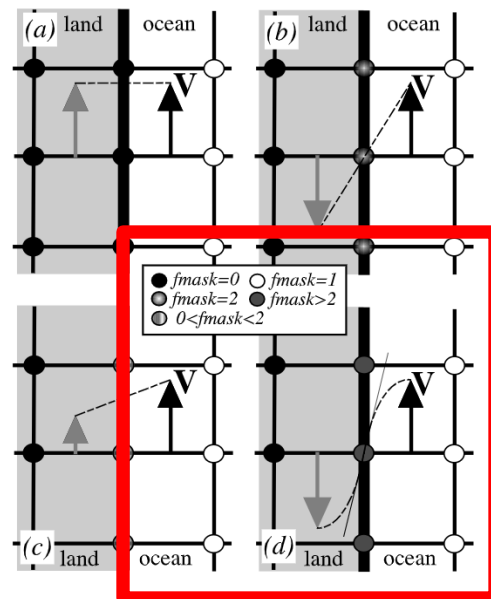


Implementation in Ocean models

”strong” no-slip boundary condition ($2 < mask-f$)

the viscous boundary layer is assumed to be smaller than half the grid size. The friction is thus larger than in the no-slip case.

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$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - v^x}{\Delta x}$$

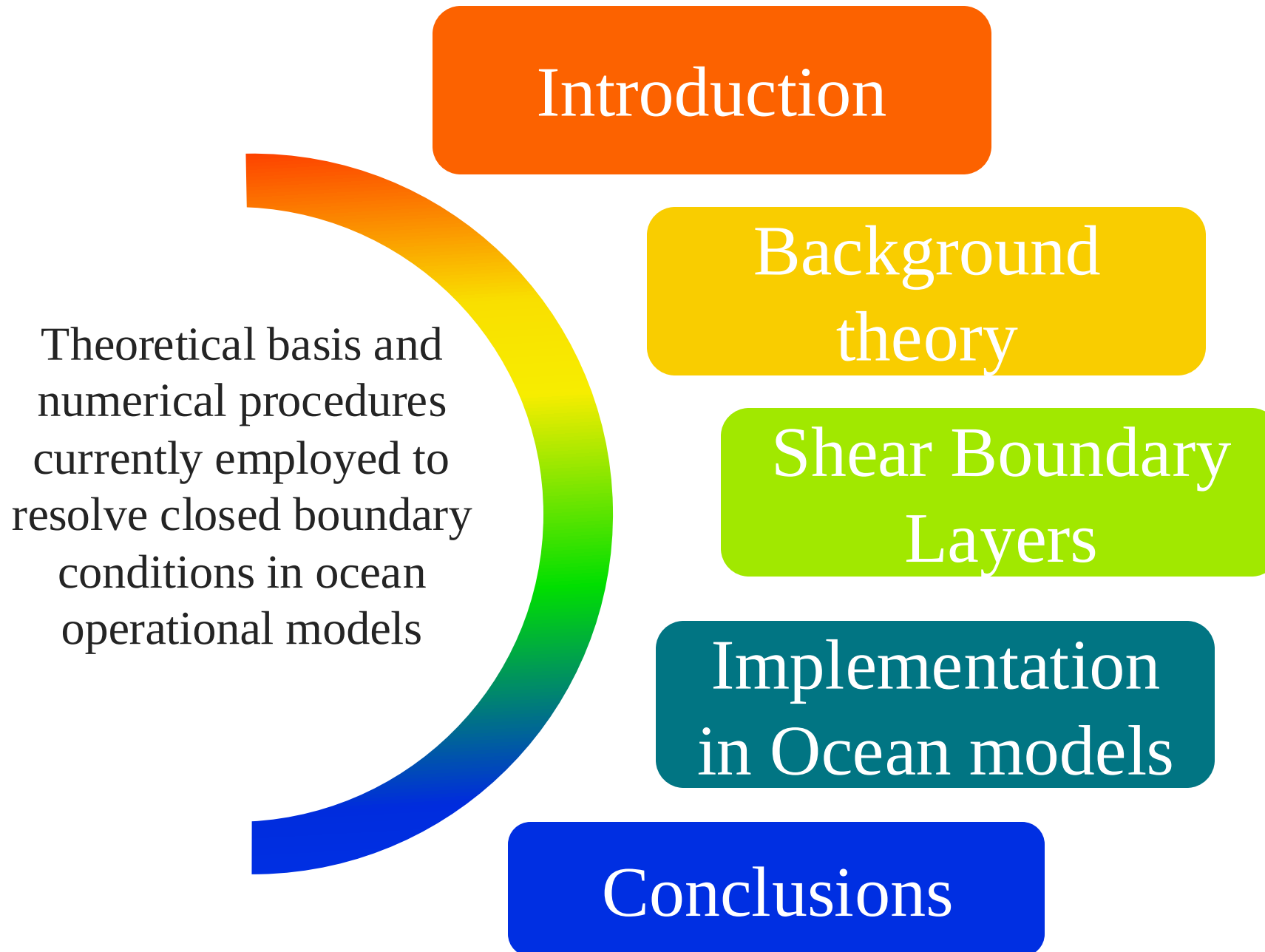
$$v^x = -\alpha v^{x+1}$$

$$\frac{\partial v}{\partial x} = \beta \frac{v^{x+1}}{\Delta x}$$

$$\frac{\partial v}{\partial x} = \frac{v^{x+1} - \cancel{v^x}}{\Delta x} \cdot mask_f$$

$$2 < mask_f$$





Conclusions



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- Navier Stokes equations require boundary conditions
- No ambiguities on the normal flow
- Considering the full field (mean + departures) the physical lateral boundary condition is the no-slip
- However, when dealing with RANS it is not so clear if no-slip/free-slip or superslip boundary condition should be applied.
- The choice has consequences on the dynamical part (in term of vorticity source/sink)
- The solution “maybe” problem dependent.
- In ocean models all the options are usually available.
- The implementation rely of the staggered nature of the grid and it is easy to implement.



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